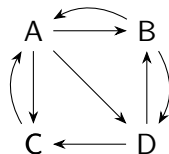


PageRank (Part 2)

Hoa Vu

Naive PageRank

We are going in to fix issues with naive PageRank.



The transition matrix with $A = 1$, $B = 2$, $C = 3$, $D = 4$ is

$$M = \begin{bmatrix} 0 & 1/2 & 1 & 0 \\ 1/3 & 0 & 0 & 1/2 \\ 1/3 & 0 & 0 & 1/2 \\ 1/3 & 1/2 & 0 & 0 \end{bmatrix}.$$

where $M_{ij} = 1/d_j$ if there is an edge from j to i and 0 otherwise.

Naive PageRank

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Suppose the surfer is at a random page at time 0. Then the distribution $\pi^{(0)} = [1/4, 1/4, 1/4, 1/4]$.

Naive PageRank

The distribution at time 1 is $\pi^{(1)} = M\pi^{(0)}$.

$$\pi^{(1)} = \begin{bmatrix} 0 & 1/2 & 1 & 0 \\ 1/3 & 0 & 0 & 1/2 \\ 1/3 & 0 & 0 & 1/2 \\ 1/3 & 1/2 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1/4 \\ 1/4 \\ 1/4 \\ 1/4 \end{bmatrix} = \begin{bmatrix} 3/8 \\ 5/24 \\ 5/24 \\ 5/24 \end{bmatrix} \approx \begin{bmatrix} 0.375 \\ 0.208 \\ 0.208 \\ 0.208 \end{bmatrix}.$$

Naive PageRank

The distribution at time 2 is $\pi^{(2)} = M\pi^{(1)} = M^2\pi^{(0)}$.

$$\pi^{(2)} = \begin{bmatrix} 0 & 1/2 & 1 & 0 \\ 1/3 & 0 & 0 & 1/2 \\ 1/3 & 0 & 0 & 1/2 \\ 1/3 & 1/2 & 0 & 0 \end{bmatrix}^2 \begin{bmatrix} 1/4 \\ 1/4 \\ 1/4 \\ 1/4 \end{bmatrix} = \begin{bmatrix} 15/48 \\ 11/48 \\ 11/48 \\ 11/48 \end{bmatrix} \approx \begin{bmatrix} 0.313 \\ 0.229 \\ 0.229 \\ 0.229 \end{bmatrix}.$$

Naive PageRank

The distribution at time 3 is $\pi^{(3)} = M\pi^{(2)} = M^3\pi^{(0)}$.

$$\pi^{(3)} = \begin{bmatrix} 0 & 1/2 & 1 & 0 \\ 1/3 & 0 & 0 & 1/2 \\ 1/3 & 0 & 0 & 1/2 \\ 1/3 & 1/2 & 0 & 0 \end{bmatrix}^3 \begin{bmatrix} 1/4 \\ 1/4 \\ 1/4 \\ 1/4 \end{bmatrix} = \begin{bmatrix} 11/32 \\ 7/32 \\ 7/32 \\ 7/32 \end{bmatrix} \approx \begin{bmatrix} 0.344 \\ 0.219 \\ 0.219 \\ 0.219 \end{bmatrix}.$$

Naive PageRank

The distribution at time 4 is $\pi^{(4)} = M^4 \pi^{(0)}$.

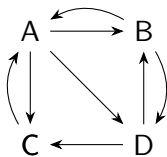
$$\pi^{(4)} = \begin{bmatrix} 0 & 1/2 & 1 & 0 \\ 1/3 & 0 & 0 & 1/2 \\ 1/3 & 0 & 0 & 1/2 \\ 1/3 & 1/2 & 0 & 0 \end{bmatrix}^4 \begin{bmatrix} 1/4 \\ 1/4 \\ 1/4 \\ 1/4 \end{bmatrix} = \begin{bmatrix} 63/192 \\ 43/192 \\ 43/192 \\ 43/192 \end{bmatrix} \approx \begin{bmatrix} 0.328 \\ 0.224 \\ 0.224 \\ 0.224 \end{bmatrix}.$$

Naive PageRank

The distribution approaches a stationary distribution π such that $\pi = M\pi$.

This gives the same definition of PageRank as before.

Naive PageRank



The transition matrix with $A = 1, B = 2, C = 3, D = 4$ is

$$M = \begin{bmatrix} 0 & 1/2 & 1 & 0 \\ 1/3 & 0 & 0 & 1/2 \\ 1/3 & 0 & 0 & 1/2 \\ 1/3 & 1/2 & 0 & 0 \end{bmatrix}.$$

$M\pi = \pi$ gives $\pi_A = \pi_B/2 + \pi_C$, $\pi_B = \pi_A/3 + \pi_D/2$,
 $\pi_C = \pi_A/3 + \pi_D/2$, and $\pi_D = \pi_A/3 + \pi_B/2$.

Naive PageRank

The distribution approaches a stationary distribution π such that $\pi = M\pi$.

From Markov process theory, such π exists and is unique if the following conditions are satisfied:

Naive PageRank

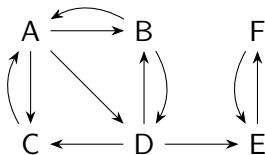
The distribution approaches a stationary distribution π such that $\pi = M\pi$.

From Markov process theory, such π exists and is unique if the following conditions are satisfied:

- The graph is strongly connected.

- The graph has no dangling nodes (dead ends).

Spider traps



The graph is strongly connected and has no dangling nodes. However, there is a spider trap. Eventually, a random surfer will get stuck in E and F .

The pagerank will be $1/2$ for E and F and 0 for the rest (which is not reasonable).

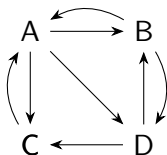
PageRank

Simplest way to make sure that the graph is strongly connected, no dangling nodes, and avoid spider traps?

Solution: allow the random surfer to jump to a page uniformly at random with probability ϵ (Google sets $\epsilon \approx 1/7$).

$$\pi_v = \sum_{u \rightarrow v \in E} \frac{\pi_u}{d_u} + \epsilon \frac{1}{n}.$$

PageRank



The new transition matrix looks like this e is the matrix of all 1's:

$$\hat{M} = (1 - \epsilon)M + \epsilon \frac{1}{n}e.$$

$$\hat{M} = (1 - \epsilon) \begin{bmatrix} 0 & 1/2 & 1 & 0 \\ 1/3 & 0 & 0 & 1/2 \\ 1/3 & 0 & 0 & 1/2 \\ 1/3 & 1/2 & 0 & 0 \end{bmatrix} + \epsilon \begin{bmatrix} 1/4 & 1/4 & 1/4 & 1/4 \\ 1/4 & 1/4 & 1/4 & 1/4 \\ 1/4 & 1/4 & 1/4 & 1/4 \\ 1/4 & 1/4 & 1/4 & 1/4 \end{bmatrix}$$

PageRank

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If we want to find the stationary distribution, we need to solve the equation $\pi = \hat{M}\pi$.

PageRank

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If we want to find the stationary distribution, we need to solve the equation $\pi = \hat{M}\pi$.

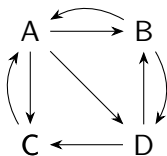
In this case:

$$\pi_A = (1 - \epsilon)(\pi_B/2 + \pi_C) + \epsilon/4.$$

$$\pi_B = (1 - \epsilon)(\pi_A/3 + \pi_D/2) + \epsilon/4.$$

...

PageRank



Suppose $\epsilon = 0.12$.

$$\pi_A = 0.88(\pi_B/2 + \pi_C) + 0.03$$

$$\pi_B = 0.88(\pi_A/3 + \pi_D/2) + 0.03$$

$$\pi_C = 0.88(\pi_A/3 + \pi_D/2) + 0.03$$

$$\pi_D = 0.88(\pi_A/3 + \pi_B/2) + 0.03.$$

Solving this gives $\pi_A \approx 0.326$, $\pi_B \approx 0.224$, $\pi_C \approx 0.224$, and $\pi_D \approx 0.224$.

Computing PageRank

Using Gaussian elimination $O(n^3)$ to solve the system of linear equations is not practical. Number of web pages is at least 1 billion.

Computing PageRank

We compute it iteratively (the power method).

Start with a random distribution $\pi^{(0)} = (1/n, 1/n, \dots, 1/n)$.

For $t = 1, 2, \dots$, update $\pi^{(t)} = \hat{M}\pi^{(t-1)}$. Equivalently,

$$\pi_v^{(t)} = (1 - \epsilon) \sum_{u \rightarrow v \in E} \frac{\pi_u^{(t-1)}}{d_u} + \epsilon \frac{1}{n}.$$

The distribution $\pi^{(t)}$ converges to the stationary distribution π .

The running time is $O(m + n)$ per iteration where m is the number of edges.

Computing PageRank

How many iterations do we need?

We define the error at iteration t as

$$\text{error}(t) = \sum_v |\pi_v^{(t)} - \pi_v|.$$

Since π is the true PageRank vector:

$$\pi_v = (1 - \epsilon) \sum_{u \rightarrow v \in E} \frac{\pi_u}{d_u} + \epsilon \frac{1}{n}.$$

Subtract this from the equation for $\pi_v^{(t)}$ to get

$$\pi_v^{(t)} - \pi_v = (1 - \epsilon) \sum_{u \rightarrow v \in E} \frac{\pi_u^{(t-1)}}{d_u} - (1 - \epsilon) \sum_{u \rightarrow v \in E} \frac{\pi_u}{d_u}.$$

Computing PageRank

Subtract this from the equation for $\pi_v^{(t)}$ to get

$$\pi_v^{(t)} - \pi_v = (1 - \epsilon) \left(\sum_{u \rightarrow v \in E} \frac{\pi_u^{(t-1)}}{d_u} - \sum_{u \rightarrow v \in E} \frac{\pi_u}{d_u} \right).$$

From the triangle inequality, we have

$$|\pi_v^{(t)} - \pi_v| \leq (1 - \epsilon) \sum_{u \rightarrow v \in E} \left| \frac{\pi_u^{(t-1)} - \pi_u}{d_u} \right|.$$

Computing PageRank

From the triangle inequality, we have

$$|\pi_v^{(t)} - \pi_v| \leq (1 - \epsilon) \sum_{u \rightarrow v \in E} \left| \frac{\pi_u^{(t-1)} - \pi_u}{d_u} \right|.$$

Summing over all v gives

$$\begin{aligned} \text{error}(t) &\leq (1 - \epsilon) \sum_v \sum_{u \rightarrow v \in E} \left| \frac{\pi_u^{(t-1)} - \pi_u}{d_u} \right| \\ &= (1 - \epsilon) \sum_v \sum_{v \rightarrow u \in E} \left| \frac{\pi_v^{(t-1)} - \pi_v}{d_v} \right| \\ &= (1 - \epsilon) \sum_v \left| \pi_v^{(t-1)} - \pi_v \right| = (1 - \epsilon) \text{error}(t - 1). \end{aligned}$$

Computing PageRank

Thus, $\text{error}(t) \leq (1 - \epsilon)^t \text{error}(0)$.

Since $\text{error}(0) \leq n$, if we want $\text{error}(t) \leq \delta n$ (i.e., the average page rank is within δ of the true value), set

$$t \geq \log_{1-\epsilon}(\delta).$$

For example, if $\epsilon = 0.12$ and $\delta = 0.001$, we need $t < 55$ iterations.

In general, the running time is $O((m + n) \log_{1-\epsilon}(\delta))$.

Using PageRank in search engine

Each search engine has its own secret sauce.

Normally, a page must have at least one keyword that matches the search query.

Other properties and PageRank are combined to rank the pages.